

Day -1 : Introduction to Networking and OSI Model Theory

1. Wireshark Capture & Protocol Names

Steps to run:

1. Open Wireshark, select your active network interface (e.g., **Wi-Fi** or **Ethernet**), and click the blue shark fin icon to start capturing.
2. Open your web browser and navigate to https://www.instagram.com.
3. Switch back to Wireshark and click the red stop button.

Protocols you will commonly see in the Protocol column:

- **DNS** (Domain Name System) - *used to find Instagram's IP address.*
- **TCP** (Transmission Control Protocol) - *used to establish connections.*
- **TLSv1.2** or **TLSv1.3** (Transport Layer Security / HTTPS) - *used for secure encrypted communication.*
- **QUIC** / **UDP** - *often used by modern browsers/apps for faster loading.*
- **IPv4** / **IPv6** - *handles network addressing.*
- **ARP** (Address Resolution Protocol) - *resolves IP addresses to local MAC addresses.*

2. OSI Layer Mapping for 3 Sample Packets

Below are 3 example packets chosen from a web browsing capture:

1. Packet 1: TLSv1.3 (or HTTP/HTTPS)

- **Protocol Name:** TLS / HTTPS
- **OSI Layer:** Application Layer (Layer 7) / Presentation Layer (Layer 6)
- **Explanation:** Handles web browsing data, encryption, and displaying content from Instagram.

2. Packet 2: TCP (SYN / ACK handshake)

- **Protocol Name:** TCP
- **OSI Layer:** Transport Layer (Layer 4)
- **Explanation:** Responsible for establishing a reliable connection, managing port numbers (e.g., Port 443 for HTTPS), and ensuring data packet delivery.

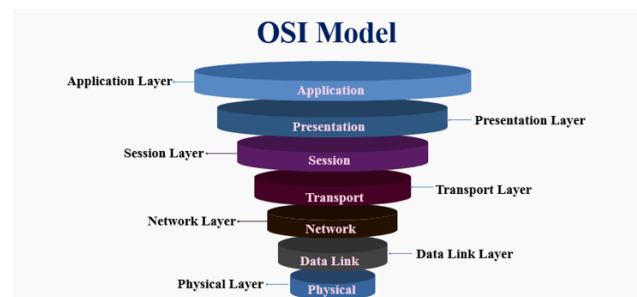
3. Packet 3: IPv4 / IPv6

- **Protocol Name:** IP (Internet Protocol)
- **OSI Layer:** Network Layer (Layer 3)
- **Explanation:** Handles logical IP addressing (source and destination IP addresses) and routing packets across the internet.

3. OSI Model Image and Real World Example

What is OSI Model

- OSI model stands for Open System Interconnection. It's a reference model which describes that how different applications will communicate to each other over the computer network.



Layer Number	Layer Name	Real-World App Example
Layer 7	Application	WhatsApp / Instagram (The app interface where you type messages or post photos).
Layer 6	Presentation	JPEG / PNG / SSL Encryption (Formats Instagram photos or encrypts WhatsApp chats).
Layer 5	Session	Zoom / Spotify Calling (Maintains active video/audio call sessions between devices).
Layer 4	Transport	TCP / UDP (Ensures WhatsApp text messages are delivered reliably and in order).

Layer Number	Layer Name	Real-World App Example
Layer 3	Network	IP Routing / Routers (Finds the best path over the internet using IP addresses).
Layer 2	Data Link	Wi-Fi / Ethernet / MAC Address (Transfers data locally between your phone and home Wi-Fi router).
Layer 1	Physical	Fiber-optic cables / Radio Waves (The actual physical signals/waves transmitting 0s and 1s).

4. How a WhatsApp Message Travels Through 4 Layers of the OSI Model

When you type *"Hey!"* on WhatsApp and tap Send, your message travels down through the OSI layers on your phone and back up on your friend's phone:

1. Application Layer (Layer 7): Typing the Message

- **What happens:** You open the WhatsApp app, type *"Hey!"*, and click the Send button. The app packages your message into a format ready for transmission.

2. Presentation Layer (Layer 6): Encrypting the Message

- **What happens:** WhatsApp applies end-to-end encryption. Your plain text message *"Hey!"* is scrambled into unreadable secret code so nobody in the middle can read it.

3. Transport Layer (Layer 4): Packaging & Reliability

- **What happens:** The encrypted message is broken into small chunks (packets). TCP adds port numbers and sequence numbers to make sure the message arrives intact without missing parts.

4. Network Layer (Layer 3): Addressing & Routing

- **What happens:** An IP address stamp is attached to the packet (your IP address as the sender, and WhatsApp server/recipient IP as the destination). Routers across the internet use this to deliver your message to the right destination